DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture Week 8: Assignment

Itô's Lemma · Geometric Brownian Motion · Lognormal Prices

Instructions

- This assignment covers Lectures 22–24 and the Week 8 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script; submit your code alongside your written answers.
- You may use software or tables for normal probabilities; state which you used.
- Submit a single PDF plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Concepts (15 marks)

A1. **(4 marks)** Explain why the ordinary chain rule fails for functions of Brownian motion. Your answer should refer to the relative sizes of ΔW and $(\Delta W)^2$.

A2. **(4 marks)** Write out the stochastic "multiplication table" for $\mathrm{d}t$ and $\mathrm{d}W$, and explain briefly where each entry comes from.

A3. **(4 marks)** Give two distinct reasons why arithmetic Brownian motion is unsuitable as a model for stock prices, and explain how GBM fixes each.

A4. **(3 marks)** True or false, with justification: "Under GBM with $\mu > 0$, the median stock price must increase over time."

Part B – Itô's Lemma (20 marks)

B1. **(5 marks)** Apply Itô's Lemma to $f(W) = W^4$ and write $\mathrm{d}(W_t^4)$ explicitly.

B2. **(6 marks)** Apply Itô's Lemma to $f(W,t) = e^{\alpha t + \beta W}$ for constants α, β. For what relationship between α and β is the resulting process a martingale? (Hint: a process is a martingale when its $\mathrm{d}t$ term vanishes.)

B3. **(5 marks)** Let $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ with μ, σ constant (arithmetic Brownian motion). Use Itô's Lemma to find $\mathrm{d}(e^{X_t})$, and identify the drift of the resulting process.

B4. **(4 marks)** Show that $\displaystyle\int_0^T W_t\,\mathrm{d}W_t = \tfrac{1}{2}\big(W_T^2 - T\big)$, and explain why the naive answer $\tfrac{1}{2}W_T^2$ cannot be correct. (Hint: consider the expectation of each side.)

Part C – Geometric Brownian Motion (25 marks)

Use the following for C1–C3: $S_0 = 250$, $\mu = 9\%$, $\sigma = 28\%$, $T = 3$ years.

C1. **(8 marks)** Compute $\mathbb{E}[S_T]$, the median of S_T, and the mode of S_T. Verify the ordering mode $<$ median $<$ mean, and compute the ratio mean/median.

C2. **(5 marks)** State the distribution of $\ln(S_T/S_0)$ completely (both parameters), and compute its standard deviation.

C3. **(5 marks)** Compute the standard deviation of S_T itself, using $\mathrm{Var}(S_T) = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1)$.

C4. **(7 marks)** Starting from the GBM SDE, derive the solution $S_t = S_0 \exp[(\mu - \frac{\sigma^2}{2})t + \sigma W_t]$ by applying Itô's Lemma to $\ln S_t$. Show every step, including the cancellation of S_t.

Part D – Lognormal Prices (20 marks)

Continue with the parameters of Part C.

D1. **(6 marks)** Compute the 5th, 50th, and 95th percentiles of S_T. Is the resulting 90% interval symmetric about the median? Explain, referencing what symmetry *does* hold.

D2. **(5 marks)** Compute $\mathbb{P}(S_T > S_0)$ and $\mathbb{P}\big(S_T > \mathbb{E}[S_T]\big)$. Comment on why the second is less than $\frac{1}{2}$.

D3. **(4 marks)** A stock has $\mu = 5\%$. Find the critical volatility above which the *median* price declines over time despite the positive expected return.

D4. **(5 marks)** Explain the long-horizon paradox in your own words: how can $\mathbb{E}[S_T] \to \infty$ while the median $\to 0$? Reference the shape of the lognormal distribution in your answer.

Part E – Computational (15 marks)

E1. **(8 marks) [Python]** Simulate 200,000 GBM paths with $S_0 = 100$, $\mu = 10\%$, $\sigma = 35\%$, $T = 4$. Report the simulated mean, median, and standard deviation of S_T alongside their theoretical values. Then repeat the simulation *omitting* the $-\sigma^2/2$ term, and report by what factor the simulated mean is inflated. Confirm this factor equals $e^{\sigma^2 T/2}$.

E2. **(7 marks) [Python]** A stock has $\sigma = 32\%$. Compute and tabulate $\mathrm{SE}(\hat{\mu}) = \sigma/\sqrt{\text{years}}$ for histories of 3, 5, 10, and 25 years, and $\mathrm{SE}(\hat{\sigma}) = \sigma/\sqrt{2n}$ for 5 years of daily data. In 2–3 sentences, comment on what length of history would be needed for $\mathrm{SE}(\hat{\mu})$ to fall below 2%, and whether that is realistic in practice.

Part F – Integrative Question (5 marks)

F1. **(5 marks)** In Week 2 you learned that a log-utility investor maximizes $\mathbb{E}[\ln W_T]$. In Week 8 you learned that under GBM, $\mathbb{E}[\ln(S_T/S_0)] = (\mu - \sigma^2/2)T$. In a short paragraph (100–150 words), connect these two facts: explain what quantity a log-utility investor is really maximizing when choosing between GBM assets, why this makes them naturally penalize volatility, and why such an investor would be unmoved by an asset with a high μ but an even higher $\sigma^2/2$.

End of Assignment. Solutions will be released after the submission deadline.